

Bachelor in Computer Applications

Course Title: **Software Engineering**

Code No: Semester: **5**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Explain incremental delivery approach to software development with a neat diagram.
2. Explain different architectural views. What are the fundamental questions that needs to be answered when designing a system architecture?
3. What do you understand by algorithmic cost modeling? Explain COCOMO estimation models.
4. What is version management? How is it carried Out?
5. Explain program inspection process.
6. Briefly explain fundamental activities related to project management.
7. Draw use case diagram for online voting system where the students can vote for their favorite teachers based on their performance. The teachers will be notified by the system about their status based on their votes. The system will generate the class routine by selecting teachers with respect to their votes. The routine will be sent to director, admin department, teachers, students and parents.
8. Explain different requirements elicitation techniques.
9. Differentiate between black box testing and white box testing.
10. Highlight on different issues that play a key role in affecting different types of software.
11. Explain the need of Model-driven architecture transformations and how is it carried out?
12. Write short notes on:
a. Agile software development
b. Context model